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Operational Insights and Key Lessons Learned from the Drilling Campaign of the First Salt Cavern Underground Oil Storage Project in the Middle East

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Abstract

This paper presents operational insights and lessons learned from the drilling campaign of the Middle East's first underground salt cavern oil storage project. The operator (Asset Owner) embarked on a pioneering program to develop a strategic crude oil reserve in deep salt caverns, marking the first implementation of this storage method in the region.

Drilling these cavern wells required overcoming unique challenges: penetrating a thick Post Hormuz Series (Cap rock) formation in a challenging geological setting, acquiring continuous full-depth core samples for analysis, controlling salt formation behavior (creep and dissolution), and ensuring absolute containment integrity to protect the environment. A comprehensive drilling and coring strategy was employed, including the use of specialised salt-saturated drilling fluids, robust well design with multiple casing barriers, and an extensive coring program that achieved 100% core recovery with a new benchmark in oriented core sector the industry overall.

The campaign was completed safely with no major incidents, and all wells were drilled to target depth with gauge wellbores, providing ideal conditions for subsequent cavern leaching operations. Key lessons include the critical importance of thorough pre-project geological analysis, real-time monitoring and adaptive drilling practices in salt, and cross-disciplinary collaboration between drilling, geology and reservoir teams. The successes and learnings from this landmark project not only demonstrate the viability of salt cavern storage in the Middle East but also offer a reference model of best practices for similar projects worldwide. The paper concludes with recommendations to guide future salt cavern drilling campaigns, emphasizing how a human-centric, innovative engineering approach can deliver both performance and safety in challenging new applications.

Introduction

Storing crude oil in underground salt caverns is a proven strategy for strategic reserves, offering secure, large-volume storage with inherent safety and cost benefits globally, salt cavern storage has been used for

decades (for example, in the U.S. Strategic Petroleum Reserve and other sites) because salt formations provide natural containment their impermeability and self-healing properties virtually eliminate leakage risk

Caverns in halite also have a lower surface footprint and environmental impact compared to conventional above-ground storage tanks (ASTs) Unlike massive tank farms that occupy large land areas and carry higher spill risks, which support the country long term sustainability strategic plan, underground caverns keep oil isolated deep underground, reducing the chance of soil or water contamination. These natural advantages, combined with salt's structural stability, make caverns an ideal choice for long-term oil stockpiling compared to conventional above-ground storage tanks (ASTs).

Until recently, however, this technology had never been implemented in the Middle East, despite the region's vast oil industry. The project discussed in this paper represents the first attempt to construct and operate salt caverns for oil storage in the Middle East and North Africa (MENA) region. The operator undertook this ambitious project as part of a strategic initiative to reinforce energy security. The facility, located at a coastal industrial terminal in the UAE, is targeting the construction of two salt caverns, each with a total capacity of approximately 1-1.5 million barrels of crude oil. The main objective of this pilot project is to thoroughly assess the salt dome area and evaluate the feasibility of scaling up underground oil storage projects in the region in the future. Each cavern is accessed via two dedicated adjacent wells (with 10m slot to slot spacing), which are drilled and completed to allow water injection and oil withdrawal. for cavern creation and operation. Implementing a salt cavern storage solution in this context introduced a host of challenges and learning opportunities.

The geological setting of salt formation (halite) overlain by different sedimentary layers required careful planning in well design and drilling practices. Traditional drilling experience in the region had to be augmented with expertise from salt cavern projects elsewhere, adapted to local conditions. Furthermore, because this was a first-in-region endeavour, an extensive data acquisition program was executed to mitigate uncertainty: this included continuous coring of the wellbores to collect complete rock samples and exhaustive formation evaluation. The operator's goals were not only to successfully drill the wells safely and on schedule, but also to capture as much insight as possible (geological, geomechanically and operational) for use in current and future possible scalability cavern projects. In short, the drilling campaign had twin aims: construct the requisite wells for the cavern storage facility and learn everything the rocks could teach to inform cavern design and risk management.

This paper details the key operational insights and lessons from the drilling campaign. In the sections that follow, we outline the project background and drilling strategy, discuss the main technical challenges encountered and the mitigation measures applied, and highlight the opportunities and successes achieved (such as record-breaking core recovery and effective risk avoidance). Finally, we will highlight the lessons learned into recommendations for future projects. The intent is to provide a comprehensive case study for industry practitioners, demonstrating how a well-planned and innovative approach can overcome the unknowns of a new application in this case, drilling for salt cavern storage and deliver a successful outcome.

Project Overview and Drilling Campaign

The underground storage project is sited at a coastal oil terminal where a massive salt dome underlies the area. The pilot phase called for two caverns to be solution-mined within the salt as forecasted in -2-3 years' time each approximately 200-300 meters in height and 60–80 meters in diameter (typical dimensions for a ~1-1.5 million expected oil storage per barrel cavern). To create each cavern, a well is drilled from surface down into the salt formation; fresh water will later be injected to dissolve the salt and form the cavern, with oil eventually stored in the resulting void. The well design for these cavern wells had to balance standard drilling requirements with the unique needs of cavern leaching and long-term storage integrity

Figure 1 schematically shows the well architecture: a large-diameter borehole is drilled and cased through the overburden formations and set just +/-150 ft inside the salt section, and then the salt section is cored

totally in main leaching well (and drilled in 8 1/2in adjacent well) into the depth range planned for the cavern. The salt section is left mostly open (uncased) to serve as the cavern space, but with an isolation casing string cemented off bottom to protect overlying strata. Each well completed by leaching strings (dual open-end concentric casing strings to allow controlled injection and production of fluids during cavern leaching and oil drawdown operations, while maintaining isolation from the surface).



Figure 1—illustrates well architecture

To execute this design, the operator launched a drilling campaign in which to total of 4 wells (two wells with 10m slot to slot spacing per each pad) these were drilled in sequence well 1 to well 4 over the project timeline (with some overlap in operations where possible). The campaign mainly spanned exploration and a foundation of possible future development phases:

- **Pilot Hole and Data Well:**

Initially, a pilot hole was drilled into the salt formation to gather critical data. Given the lack of regional experience with salt geology, this first well was designed to be fully cored and extensively logged. The well was spudded with a standard surface hole 8 1/2" coring assembly from surface, but due to the excessive surface hole losses the surface coring terminated after excessive trials to control the surface losses, instead 26" diameter) drilled to a depth of about 800 m, where 18 5/8" surface casing was set to isolate unconsolidated near-surface strata and any aquifers. Below that, the excessive coring program to core all salt sections in well 1 and well 3 utilizing advanced oriented coring technique, the cored section acquired in 8 1/2" x 41/2" section followed by excessive open hole logs across the salt formation, retrieving core samples all the way to the base of the salt interval (which extended to approximately 3,500 ft depth in this location). This yielded a complete column of core representing the entire salt dome section and surrounding formations. The coring operation was a centrepiece of the pilot well plan, as it allowed direct examination of the

salt's purity, mechanical properties, and any interbedded layers. Remarkably, the coring program achieved 100% recovery of core over the salt section

It is worth mentioning that the core planning stage lasted over a year in order to finalize and agree on the game plan. As a result the coring program not only achieved 100% recovery of core over the salt section but also beat the world know record or oriented core in single run three times in row and set a new benchmark of 361 ft of oriented core in single run.

This was accomplished by employing a specialised salt-saturated mud system (to prevent salt dissolution and preserve core integrity) and a tight quality control on core handling at surface (to avoid fragmentation or contamination of the samples). The success of this pilot/data well provided the team with invaluable geological and geomechanically information, reducing uncertainty for the subsequent cavern wells.

- **Cavern Wells:**

With data in hand, the project proceeded to drill the remaining cavern wells (and deepen the pilot well to final specifications if needed) that would become the actual storage caverns. These wells followed a similar design mentioned and highlighted in [figure 1](#), Throughout the drilling campaign, real-time data acquisition played a vital role. Measurement-While-Drilling (MWD) tools provided continuous inclination and azimuth measurements to ensure the wells remained vertical. In the pilot hole, a comprehensive wireline logging suite followed the coring, including sonic logs, formation imaging, and even specialty logs to assess salt quality (as geo-radar which used for the fist time this region to evaluate extent of salt section). The drilling team was supported by a dedicated remote operations centre that monitored parameters for any signs of trouble such as stuck pipe tendencies or influx/losses. Fortunately, no major drilling problems occurred no influxes, no significant losses (across the salt section), and no stuck pipe incidents were recorded in any of the wells, a testament to the careful planning and constant vigilance.

By the end of the campaign, all four wells were successfully drilled and cased as designed. The full-length cores and extensive log data were handed over to reservoir and geomechanics experts for cavern design calibration. With the wells in place, the project moved on progress to move to the leaching phase (beyond the scope of this paper), where fresh water is plan to inject to dissolve the salt and create the caverns.

[Figure 2](#) provides a schematic of how the cavern wells, should be after leaching illustrating how the final cavern design should be after leaching.

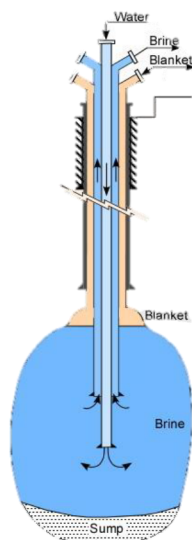


Figure 2—a schematic of cavern well after leaching.

In summary, the drilling campaign established a solid foundation both physically (in well infrastructure) and intellectually (in subsurface knowledge) for the success of this underground storage project.

Challenges and Mitigation Measures

The project encountered several technical and operational challenges during the drilling campaign, stemming largely from the unfamiliar geology and the exacting requirements of cavern storage wells. The key challenges and how they were addressed are outlined below:

- **Drilling through a Massive Salt Formation:** The salt dome presented an atypical drilling medium compared to the region's usual formations. **Halite rock** is relatively soft (compared to carbonates or sandstone) but behaves plastically under stress and heat; it can deform and "creep" into the wellbore, especially if the open hole is left unsupported for too long. Additionally, exposing salt to undersaturated water-based mud can lead to rapid dissolution and washout, enlarging the hole uncontrollably. *Mitigation:* To maintain wellbore stability in salt, the team carefully selected the drilling fluid and minimized the exposure time of open hole. A saturated salt mud system was used while drilling the salt sections, ensuring the fluid was in equilibrium with the formation and thus greatly reducing dissolution. The mud weight was set just enough to balance any minor pressures and support the hole, but not excessively high (overbalance pressure on salt can encourage creep by forcing fluid into micro-fractures). Drilling in salt was done round-the-clock with one rig in operation. These tactics prevented significant creep closure or washout post-drilling and hole enlargement operation calliper logs showed the salt holes remained near-gauge, and the casing could be run to bottom without issues. Another challenge in the salt was the presence of interbedded hard streaks (anhydrite layers) within the halite, which can cause transient drilling difficulties. The bit would drill rapidly in pure salt then suddenly slow and torque up on hitting an anhydrite layer. To mitigate this, the BHA was stabilized with near-bit reamers and shock subs to dampen vibrations, and drilling parameters were tuned (reducing weight-on-bit slightly before expected hard streaks, to avoid bit damage). Bit selection was also optimized: for instance, one well employed a hybrid bit that combined PDC cutters for salt with a few impregnated diamond elements to grind through anhydrite stringers effectively. These measures collectively addressed the salt drilling challenge, resulting in trouble-free penetration of salt in each well with minimal diameter enlargement.
- **Extensive Core Acquisition and Preservation:** Obtaining a continuous core from surface to TD (especially through the salt) was a priority for the pilot well, but it introduced complexity. Coring is inherently slower than regular drilling and involves different risks the core could jam, break, or be lost while pulling out, and the open hole is exposed for longer during coring runs. Additionally, salt cores are prone to degrading (dissolving or fracturing) upon retrieval due to pressure release and exposure to humidity. *Mitigation:* A specialized coring service was engaged, utilizing a large-diameter core barrel (8 1/2 inch core) to capture as much formation per run as possible and reduce the number of trips. To combat potential sticking of the core barrel in the salt, the coring mud was maintained at high salinity and enhanced lubricity, and the scoring runs were kept managed to enhance the well deliver performance. Each core run was monitored for penetration drops that might indicate core jamming, and none were observed. At surface, an elaborate core preservation workflow was implemented. As soon as a salt core was brought up, it was immediately coated in a specialized wax and wrapped in multiple layers of plastic film under anhydrous conditions to lock in its native state fluids and prevent moisture ingress. The core was then placed in sealed PVC tubes. This process prevented any evaporation or salt crystallization that could break the cores apart. The result was an unprecedented 100% core recovery rate and excellent core quality

The preserved cores were later analysed in detail, providing insight into rock mechanical properties (creep rates, compressive strength), lithology, and insoluble impurities in the salt. These insights fed directly into better cavern design (e.g., determining leaching rates and the need for any impurity management). A lesson learned here was that meticulous coring procedures and preservation techniques can yield laboratory-quality samples even from challenging materials like halite, which historically have low recovery due to dissolution or breakage. This has set a new benchmark for coring performance in the company's operations.

- **Wellbore Stability and Hole Integrity:** Beyond the salt itself, drilling through the overlying formations and into the competent strata below posed their own challenges. One concern was encountering any pressurized zones or weak formations above the salt that could cause kicks (influx of fluids) or lost circulation. Another was ensuring the open hole in the salt remained stable until completion. *Mitigation:* The team adopted a cautious approach with regard to pore pressure prediction and mud weight management. Offset data and regional analogy were studied to estimate pressures; while the salt is impermeable, the cap rock above it (often a carbonate or anhydrite layer at the top of salt) and any underlying formations could hold high pressures. The mud weight was gradually increased when approaching the expected top of salt, to slightly overbalance any possible pocket of trapped fluids. This incident validated the proactive strategy of isolating the formations above salt with casing as soon as the salt top was penetrated. Regarding the salt section's stability: in addition to the fluid measures described, the project planned the casing program such that the salt would be left unsupported for the shortest time possible. As soon as drilling reached the intended cavern bottom depth, operations shifted to running the cemented casing. In effect, the salt section open hole was kept exposed just for the duration of logging and then turned over to leaching.
- **Ensuring Zonal Isolation and Casing Integrity:** For a storage well, well integrity is paramount. The casing strings, particularly the surface casing, form critical barriers to prevent any leakage of oil from the cavern to shallower groundwater or to surface. Cementing across salt can be tricky salt's plastic nature and solubility can lead to micro-annuli or poor bonds if not done correctly. *Mitigation:* The project implemented rigorous cementing practices tailored for salt environments. A salt-saturated cement slurry was used for the casing set at the top of salt, to avoid the cement water dissolving the salt and causing a void. Expanding additives were included in that cement formulation to counteract any shrinkage and to ensure the cement plugs any micro gaps as the salt creeps slightly over time. Cement bond logs (CBL) were run after setting this casing. Additionally, the casing design accounted for salt load: the casing running through or just above salt was of a higher collapse grade to resist external forces if the salt tried to squeeze it. Another isolation consideration was the interface between crude oil and water in the cavern to protect the casing from corrosion. While this goes beyond drilling phase, it highlights that drilling and completion (leaching string) had to be integrated to ensure long-term integrity. During drilling, every casing shoe was tested rigorously (formation integrity tests) to confirm it could hold the required pressures for the next phase, given that later on, cavern operations would involve cycling pressures.
- **Operational Safety and HSE Considerations:** Conducting a complex drilling campaign with novel elements (like extensive coring and new mud systems) required a sharp focus on HSE. Potential hazards included handling large volumes of oil-based mud (for special logging requirement) and salt-saturated brine, managing heavy core samples and specialty equipment, and coordinating a larger-than-usual team of geologists, mud engineers, and drilling crew on the rig. *Mitigation:* The operator put in place an enhanced HSE plan for this project. This included pre-spud workshops to brief all personnel on the specific risks (for example, discussing the unusually heavy core barrels that would be pulled out and the need for proper lifting practices, or the safe handling of hot, saturated salt mud which can cause skin irritation). Additional safety personnel were assigned on the rig during critical operations like coring runs and casing installations. The mud system,

being salt-saturated, was handled with care to prevent spills all flowlines and pits were covered or enclosed to avoid splashes, and spill kits were on standby (ultimately no significant spill occurred). Overall, the campaign was executed with zero lost-time incidents (LTI) and no environmental harm, reflecting a successful safety management effort. This was a significant achievement given that many aspects were first-of-a-kind for the crew. A lesson reinforced by this experience is that extra upfront training and HSE focus are necessary when pushing technical boundaries – the crew's familiarity with standard drilling had to be expanded to cover core handling, saturated fluids, etc., and the proactive approach ensured everyone went in prepared.

In summary, each challenge in the project was met with carefully engineered solutions. By leveraging global best practices (from other salt cavern projects) and tailoring them to the local context, the team avoided pitfalls that could have arisen. This cross-project learning exemplifies how knowledge sharing can de-risk novel operations. [Table 1](#) below summarizes the major challenges vs. mitigation strategies in a concise format, demonstrating the systematic approach adopted:

Table 1—Key Challenges and Mitigations in the Cavern Drilling Campaign

Challenge	Mitigation Strategy
Drilling thick salt prone to creep and washout	- Used saturated high-density brine mud to prevent dissolution - Minimized open-hole time (drilled salt section in one go especially during the hole opening) - Optimized BHA and parameters for salt (stabilizers, shock subs for vibrations)
Obtaining full-depth core without loss or damage	- Employed longer core barrels for fewer runs - Strict core handling/preservation (wax coating, sealed containers) - High-salinity mud and prompt surface processing to prevent core degradation
Unexpected pressurized zones above salt	- Conducted thorough pre-drill pressure prediction - Increased mud weight cautiously before entering salt cap rock - Set intermediate casing immediately after salt top, if needed (as done in pilot well after gas show)
Cementing and casing integrity in salt	- Used salt-saturated, expanding cement to ensure bond - Selected high-collapse, corrosion-resistant casing for salt section - Ran cement bond logs to verify annular seal, achieved good isolation
Operational unfamiliarity (crew and logistics)	- Held specialized training for coring and salt mud handling - Augmented crew with subject-matter experts (salt geologist, mud chemist) on site - Enhanced safety protocols (extra PPE for OBM, careful lifting practices for heavy cores) resulting in zero LTIs

Opportunities and Successes

Alongside the challenges, the drilling campaign revealed several opportunities and positive outcomes that are worth highlighting:

- Validation of Salt Cavern Feasibility in the Region:** Perhaps the most significant outcome is the proof that a salt cavern storage project can be successfully executed in the Middle East. Prior to this, there was uncertainty around how the region's geological conditions (lack of local salt project experience) would affect such a project. The drilling campaign demonstrated that with the right expertise and planning, the region's salt formations can be drilled and managed effectively. This opens the door for similar projects in other suitable salt domes in the area, knowing that the technical risks can be mitigated. In broader terms, the project provides a template for developing

strategic petroleum reserves in other parts of the world that have salt but haven't utilized it for storage yet – an important contribution as energy security and strategic storage become ever more critical.

- **Strategic Energy Security and Flexibility:** On an operational level for the Asset Owner, the project once completed will yield a big strategic advantage. The caverns (once leached and filled) will provide on-demand storage that the operator can use to buffer market fluctuations. For instance, surplus production can be stored cheaply underground and quickly retrieved when needed, without the evaporation losses or security risks of surface tanks. This flexibility allows the operator to respond more efficiently to market needs

From a drilling perspective, contributing to this strategic capability was a motivating factor; it drove home the importance of getting the wells right. The success of drilling means the leaching can proceed on schedule, and the expected operational date of the storage is met, so the broader project objectives remain on track. In essence, the drilling team's achievements directly enable the country's first strategic oil reserve to materialize, which is a national milestone.

- **Environmental Benefits Realized:** As discussed, underground caverns have environmental advantages over above-ground storage. By executing this project, the operator is lowering the overall environmental footprint of their storage operations. One outcome to note is that, due to the efficient drilling and isolation, there was no contamination of groundwater or soil during the drilling phase all wells were sealed off properly, and despite dealing with oil-based fluids and salt cuttings, the team contained all materials.
- **Technological Innovation and Records:** The project spurred the use of cutting-edge technology in the operator's drilling practice. The full-depth coring with 100% recovery is one clear example it set a new company record for longest continuous core interval recovered. This not only provided great data but also demonstrated the value of investing in advanced coring techniques, which can be applied in other exploration wells needing core. The project also saw the first use of certain drilling fluids and bits in the company's operations (like the saturated salt mud). By pushing boundaries, the drilling team learned a great deal about managing complex operations (like simultaneously handling coring, extensive logging, etc.). The knowledge gained has already been captured in internal best practice documents. One could say the project served as a technology incubator for the company's drilling department giving engineers a chance to implement and refine novel approaches that can now be standardised. Moreover, achieving undoubtable achievements by beating the world record of oriented core to 361 ft per run, perform the tandem hole opening by hole opener for the first time in the region and capitalize on the lessons learned that all a big achievement that has boosted confidence. It encourages attempting other innovative drilling methods (for example, in the future, drilling maybe directional salt wells or investigating hydrogen storage in caverns, the company can leverage this positive experience). On a personnel level, the team involved has significantly upskilled; their expertise in salt drilling is now a valuable asset that can be shared across the organization.
- **Data for Geoscience and Engineering:** The wealth of data collected (cores, logs, pressure measurements) is an opportunity in itself. It has been used to refine the cavern designs. The core-derived lithology identified some clay/shale partings in the salt; knowing this, the leaching process can be tuned to handle insoluble residues (clay won't dissolve like salt, so you have to manage those fines). In essence, the drilling campaign doubled as a scientific exploration of the salt dome. It also provided calibration for seismic surveys by tying core and logs to seismic, geologists improved the subsurface model of the dome, which can guide exact cavern placement to avoid any anomalies. All these insights reduce operational risk in the next phases and ensure the caverns achieve their

desired capacity and longevity. The success of the coring and logging program hence translates to a more optimized storage facility. Furthermore, the data has value beyond this project: it's the first extensive dataset on Gulf-region salt mechanics and can be shared (in parts via technical papers like this one) to the broader industry, contributing knowledge on topics like salt creep in a Middle Eastern context or coring best practices in salt.

In summary, the drilling campaign not only met its immediate objectives (delivering wells for the cavern project) but also generated additional benefits strategic, environmental, and technical. It showcased that with rigorous engineering, innovation can go hand-in-hand with operational excellence. The project's interim successes have been celebrated within the company and set a positive tone for the remaining project phases.

Key Lessons Learned

The experience from this project yields several key lessons and best practices that can inform future drilling campaigns, especially those tackling first-of-a-kind challenges. The main lessons learned include:

1. **Thorough Upfront Planning and Multidisciplinary Collaboration are Essential:** From the start, this project integrated input from geologists, drilling engineers, reservoir and cavern design specialists, and even external experts familiar with salt projects. This holistic planning ensured that well design and drilling plans were fit-for-purpose (e.g., coring plans matched geological info needs, casing plans aligned with cavern design). Future projects should similarly employ a multidisciplinary approach early – it was instrumental in foreseeing challenges and aligning objectives across teams.
2. **Leverage Analogous Experiences but Adapt to Local Conditions:** Since this was new to the region, the team wisely sought knowledge from similar projects internationally (literature, experts, etc.)
3. **Robust Data Acquisition Pays Off:** The decision to core the entire pilot well demanded extra time and cost, but it paid dividends in risk reduction and optimized design. Having that full data set built confidence and prevented costly surprises later. The lesson for future projects (even beyond salt cavern ones) is that investing in front-end data gathering whether through coring, extended logging, or formation testing can significantly de-risk complex operations. It provided direct ground truth that complemented seismic and models, leading to better decision-making.
4. **Custom Drilling Fluid Design is Crucial in Extreme Environments:** One of the unsung heroes of this campaign was the mud system. Off-the-shelf mud would not have sufficed for drilling salt; the project's custom salt saturated system was key to maintaining wellbore stability and core quality
5. **Manage Open-Hole Exposure Time:** Especially in reactive or deformable formations like salt (or some shales), time is a critical factor the longer a hole stays open, the higher the chance of problems. This project underscored the value of time compression in drilling sequences. Drilling continuously through the salt and expediting casing running was vital to success. In operations planning, identifying such "time-sensitive" intervals and pre-planning to drill and secure them in one go is a best practice that can prevent many downhole issues.
6. **Advanced Coring Techniques Can Achieve High Performance:** Historically, achieving 100% core recovery in long intervals was uncommon. By using modern coring tools and meticulous procedures, the team showed that "perfect" core recovery is possible even in difficult formations. The lesson is that if core data is high priority, one should not shy away from coring due to old perceptions of low recovery with today's technology and proper vendor partnership, extremely high recovery and quality can be attained. This might encourage more widespread coring in projects that would benefit from core data but usually avoid it due to past poor results, [figure 3](#) show real comparison and optimization impact from longer core and its overall impact on the wells delivery and the project as whole by saving +50% of the coring time of the well1 and overall of 19.5 days of allocated business plan days of coring operation.

7. **Continuous Monitoring and Real-Time Decision Making:** This campaign benefited from having extra eyes on real-time data (both at the rig and remotely). The early detection of the gas show and prompt response is one example. Another is adjusting drilling parameters on the fly when hitting hard stringers. It reinforces that for complex operations, investing in real-time monitoring capability (and empowering the team to make adjustments) prevents incidents and optimizes performance. It's far better to catch a stick-slip event early and mitigate it than to deal with stuck pipe after the fact, for instance. Future projects should ensure strong telemetry, data visualization, and an experienced real-time support team for critical wells.
8. **Well Integrity Can't Be Overemphasized:** Given the end-use of these wells (long-term storage), the project maintained a strict focus on quality. The lesson is that when wells are meant for something beyond just drilling (like storage or injection), one must hold integrity requirements to the highest standard. Shortcuts in casing or cementing are simply not acceptable as they can compromise the entire project's viability. The meticulous approach to well integrity in this campaign should be a template for any similar projects.
9. **Document and Disseminate Lessons Quickly:** The project team made efforts to document all important observations (through daily reports, after-action reviews, etc.). These were compiled into internal knowledge base items and also form the basis of publications (including this paper and others presented at ADIPEC 2024). This ensures that the corporate memory retains these lessons and that others can learn from them, preventing reinvention of solutions. A good practice is to have a "lessons learned" session immediately after each well, while memories are fresh, and to update drilling guidelines accordingly. This was done here for example, after the pilot well, the drilling program for the next wells was updated to incorporate the core findings and minor tweaks in procedures. Future projects should follow a similar continuous improvement loop.
10. **HSE and Training for Novel Operations:** A final but crucial lesson: when undertaking novel operations, invest in extra HSE measures and training. The fact that this campaign had zero LTIs is not luck; it was the result of thorough training of personnel (like how to handle core safely, how to work with different fluid) and a proactive safety culture. Introducing new equipment or tasks inherently carries a learning curve addressing that through training and supervision is essential to avoid accidents. This project proved that with the right mindset, even first-time operations can be executed safely. As a result, the workforce is now more skilled and confident, which is an asset going forward.



Figure 3—illustrates coring wells durations

These lessons learned have been compiled and distributed within the operator's drilling and completions community to inform ongoing and future projects. Many of them are broadly applicable, reinforcing best practices in planning, execution, and safety. Implementing these lessons will help ensure that subsequent endeavours whether drilling another salt cavern or tackling any complex drilling scenario can achieve similar or even greater levels of success.

Conclusion

The drilling campaign for the Middle East's first salt cavern oil storage project was completed successfully, providing valuable insights and establishing best practices for future projects of this nature. The key conclusions and takeaways are:

- **Project Feasibility and Success:** The campaign demonstrated that drilling wells for salt cavern storage in the region is not only feasible but can be accomplished safely and efficiently. All planned wells were drilled to target depth with their objectives met (full data capture and proper construction), validating the viability of using local salt formations for strategic oil storage. This is an important proof-of-concept for the operator and the region at large.
- **Technical Innovations Enable Performance:** The use of innovative techniques notably the comprehensive continuous coring and the customized drilling fluid strategy was pivotal in overcoming the unique challenges. By applying these innovations, the team achieved record performance (such as 100% core recovery and zero significant NPT) that exceeds typical industry benchmarks for similar operations. It underlines that innovation in drilling technology and practice can unlock solutions to what might initially appear as daunting challenges (like drilling salt or obtaining long cores or even use more for less as the use of the tandem hole opening using hole openers for the first time this region).
- **Importance of Data and Analysis:** The wealth of geological and geomechanically data gathered (cores, logs, etc.) proved invaluable in guiding the project and will continue to inform the cavern development. This highlights a broader point: investing in data upfront yields returns in better design and risk mitigation. The cavern project is likely to be more successful (in terms of stability and capacity) as a direct result of the information the drilling phase provided. In future complex projects, prioritizing data acquisition is recommended as a strategy to reduce uncertainty.
- **Risk Mitigation and No-compromise on Safety:** The campaign had a strong focus on risk identification and mitigation from mud design to casing plans and HSE protocols which resulted in a smooth operation with no major incidents. This underscores the value of comprehensive risk management in drilling complex wells. Each potential problem was countered with a pre-planned solution, illustrating that upfront risk analysis and contingency planning are well worth the effort. Additionally, the project reaffirmed that safety and operational excellence go hand in hand: maintaining a safety-first approach did not slow the project; on the contrary, it likely prevented disruptions, thereby helping the project finish on time.
- **Strategic and Environmental Impact:** On a strategic level, the drilling campaign has paved the way for an important asset – a large-scale underground oil storage capability that will enhance energy security and operational flexibility for the operator and the host country. Environmentally, the project's success means that a safer, lower-footprint storage method is being realized, which has positive implications for sustainability. Another positive outcome of the technical solutions as tandem hole-opening operation was a reduction in the environmental footprint of drilling activities. Moreover, the elimination of an extra drilling time through the technological solutions would minimizing emission. This streamlined approach aligns with the operator's sustainability goals it not only saved rig time and cost but also lowered the well's overall carbon footprint. The trial thus demonstrates not only drilling efficiency gains but also how innovative practices can contribute to emissions reduction in line with industry decarbonization efforts.
- **Lessons for the Future:** The lessons learned from this campaign (as detailed in the previous section) form a valuable knowledge base. For the operator, these lessons are already influencing other projects for instance, improved coring techniques are being considered in upcoming exploration wells, and the approach to drilling fluids in problematic formations (like swelling shales) is being revisited with this project's success in mind. The industry at large can also draw

from this experience. As more regions explore underground storage (including for crude oil, natural gas, or even hydrogen in the future), the case study of this project offers a reference on how to approach the drilling aspects effectively. Sharing these findings through SPE and other forums will hopefully aid engineers elsewhere.

In closing, the drilling phase of the salt cavern project achieved all its objectives and in many cases exceeded expectations. It set new standards for the company in terms of technical achievement and provided a rich set of data and insights. Most importantly, it laid the groundwork for the successful creation of the caverns, which are on track to become operational and deliver strategic value. The authors conclude that the keys to this success were meticulous planning, openness to innovation, and rigorous execution. Future projects tackling unconventional challenges can replicate this formula: start with strong foundational designs, be willing to push the envelope with new techniques (backed by sound engineering), and maintain discipline in execution with an eye on safety and quality at every step. The Middle East's first salt cavern storage drilling campaign will stand as a reference point for years to come in what it achieved and how it achieved it. The hope is that its legacy will not only be the physical caverns but also a transfer of knowledge that spurs continued innovation in drilling and underground storage practices.

Nomenclature

MENA	: Middle East and North Africa
ASTs	: Above-ground storage tanks
SPR	: Strategic Petroleum Reserve
OHL	: Open-hole logs
WIT	: well integrity test
MIT	: Mechanical Integrity test
HSE	: Health, Safety, and Environment
NPT	: Non-Productive Time (time during operations where no progress is made, often due to issues)
Salt Creep	: Time-dependent plastic deformation of salt, which can cause the borehole to shrink in diameter
SSM	: Salt Saturated mud

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